```
# Responsible AI & Model Interpretation

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
from sklearn.inspection import permutation_importance

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
# Step 3: Load the dataset

data = load_breast_cancer()

X = pd.DataFrame(data.data, columns=data.feature_names)
y = data.target

print("Dataset loaded successfully!")
print("Number of rows:", X.shape[0])
print("Number of features:", X.shape[1])
print("\nFirst 5 rows:")
display(X.head())
```

```
Dataset loaded successfully!
Number of rows: 569
Number of features: 30

First 5 rows:
```

	mean radius	mean texture	mean perimeter	mean area	mean smoothness	mean compactness	mean concavity	c
0	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001	
1	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869	
2	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974	
3	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414	
4	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980	

5 rows × 30 columns

```
# Step 4: Split the dataset
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X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("Training samples:", X_train.shape[0])
print("Testing samples:", X_test.shape[0])

# Train the machine learning model
model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)

model.fit(X_train, y_train)

print("\nModel trained successfully!")
```

```
Training samples: 455
Testing samples: 114

Model trained successfully!
```

```
# Step 5: Evaluate the model

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Model Evaluation Results")
print("------------------------")
print("Accuracy:", round(accuracy * 100, 2), "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

```
Model Evaluation Results
------------------------
Accuracy: 95.61 %

Classification Report:
              precision    recall  f1-score   support

   malignant       0.95      0.93      0.94        42
      benign       0.96      0.97      0.97        72

    accuracy                           0.96       114
   macro avg       0.96      0.95      0.95       114
weighted avg       0.96      0.96      0.96       114
```

```
# Step 6: Feature Importance

importance = pd.DataFrame({
```

```
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

importance = importance.sort_values(
    by="Importance",
    ascending=False
)

print("Top 10 Important Features:")
display(importance.head(10))

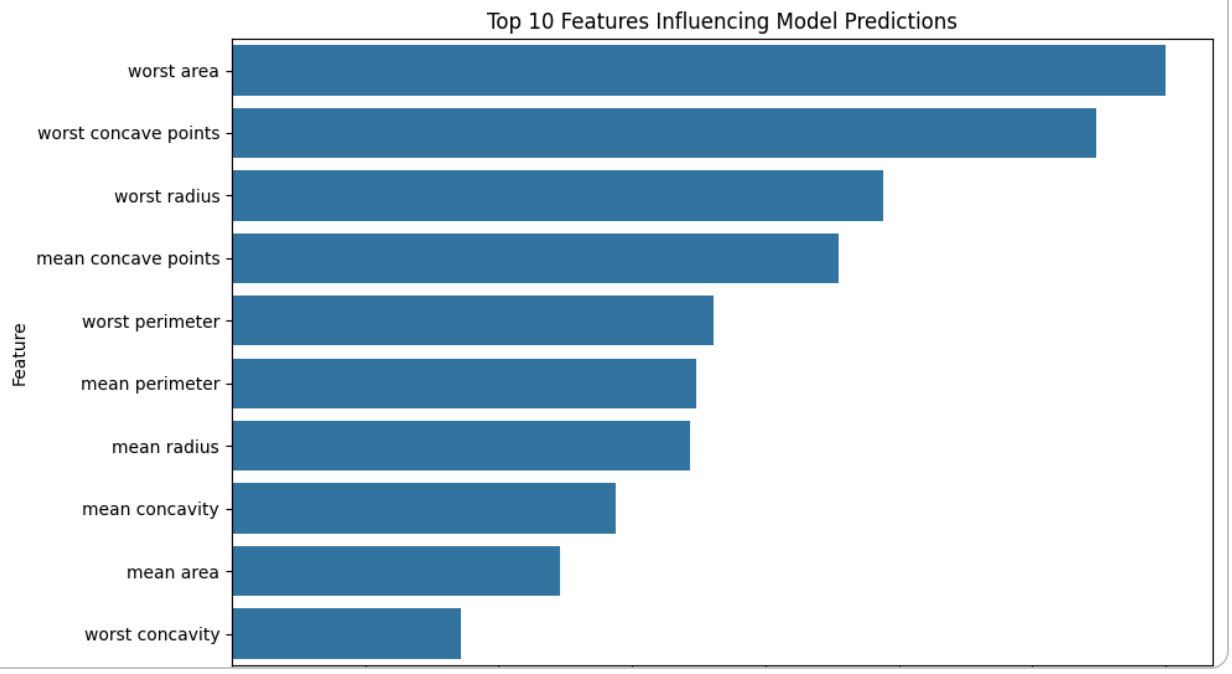
# Visualize feature importance
plt.figure(figsize=(10, 6))

sns.barplot(
    data=importance.head(10),
    x="Importance",
    y="Feature"
)

plt.title("Top 10 Features Influencing Model Predictions")
plt.xlabel("Importance Score")
plt.ylabel("Feature")
plt.tight_layout()
plt.show()
```

Top 10 Important Features:

	Feature	Importance
23	worst area	0.140016
27	worst concave points	0.129530
20	worst radius	0.097696
7	mean concave points	0.090885
22	worst perimeter	0.072226
2	mean perimeter	0.069574
0	mean radius	0.068676
6	mean concavity	0.057638
3	mean area	0.049172
26	worst concavity	0.034340




```
# Step 7: Check Model Performance Across Groups

results = pd.DataFrame({
    "Actual": y_test,
    "Predicted": y_pred
})

results["Correct_Prediction"] = (
    results["Actual"] == results["Predicted"]
)

# Compare performance for each actual target group
group_results = results.groupby("Actual").agg(
    Total_Samples=("Correct_Prediction", "count"),
    Correct_Predictions=("Correct_Prediction", "sum"),
```

```
    Accuracy=("Correct_Prediction", "mean")
).reset_index()

group_results["Group"] = group_results["Actual"].map({
    0: "Malignant",
    1: "Benign"
})

group_results["Accuracy"] = (
    group_results["Accuracy"] * 100
).round(2)

print("Performance Comparison Across Target Groups:")
display(group_results[
    ["Group", "Total_Samples", "Correct_Predictions", "Accuracy"]
])

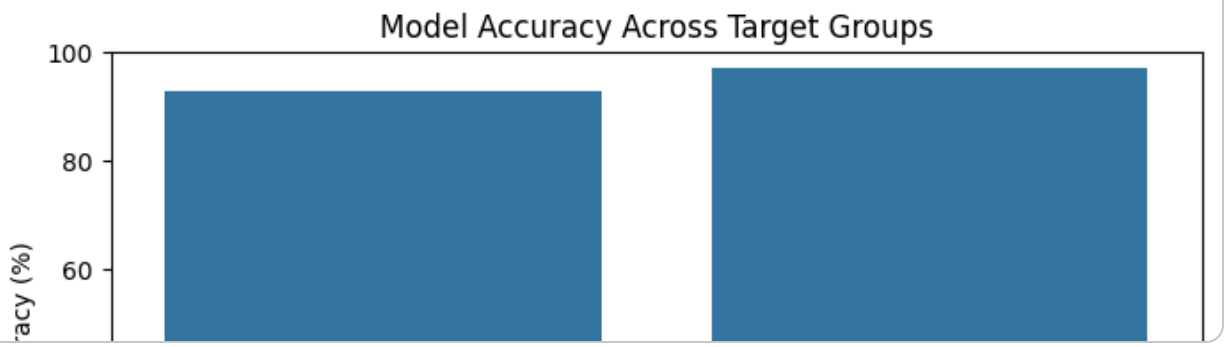
# Visualize the comparison
plt.figure(figsize=(7, 4))

sns.barplot(
    data=group_results,
    x="Group",
    y="Accuracy"
)

plt.title("Model Accuracy Across Target Groups")
plt.ylabel("Accuracy (%)")
plt.xlabel("Target Group")
plt.ylim(0, 100)
plt.tight_layout()
plt.show()
```

```
Performance Comparison Across Target Groups:
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	Group	Total_Samples	Correct_Predictions	Accuracy
0	Malignant	42	39	92.86
1	Benign	72	70	97.22




```
# Step 8: Confusion Matrix

from sklearn.metrics import confusion_matrix

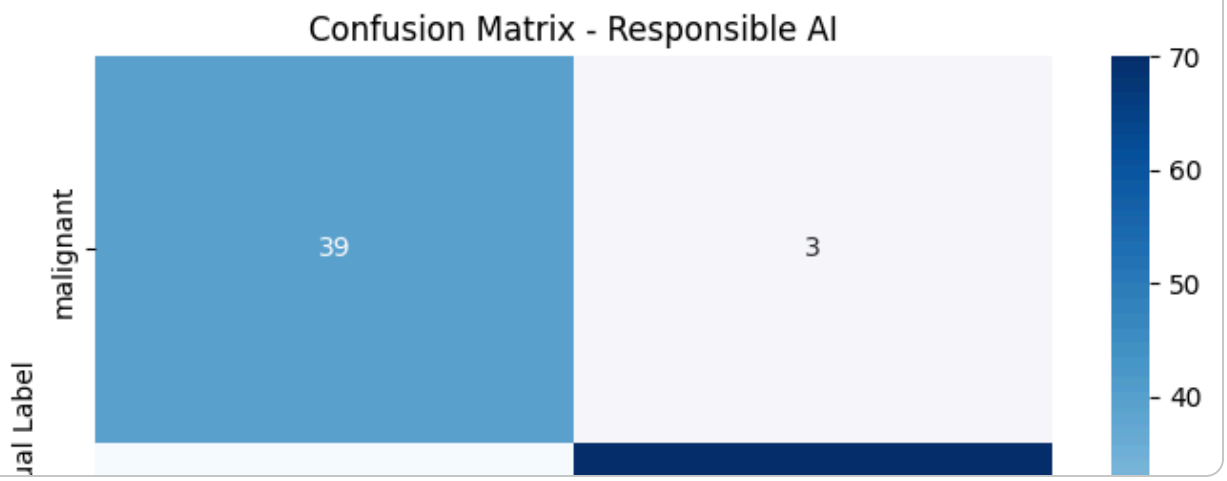
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(7, 5))
```

```
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=data.target_names,
    yticklabels=data.target_names
)

plt.title("Confusion Matrix - Responsible AI")
plt.xlabel("Predicted Label")
plt.ylabel("Actual Label")
plt.tight_layout()
plt.show()

print("Confusion matrix generated successfully!")
```




```
# Step 9: Responsible AI Report

print("=" * 55)
print("       RESPONSIBLE AI & MODEL INTERPRETATION")
print("=" * 55)

print("""
1. PROJECT OBJECTIVE
This project demonstrates machine learning model
interpretation using a Random Forest classifier.

2. MODEL USED
Random Forest Classifier.

3. DATASET
The Breast Cancer Wisconsin dataset provided by
Scikit-learn.

4. MODEL PERFORMANCE
The model was evaluated using accuracy, precision,
recall, F1-score, and a confusion matrix.

5. INTERPRETABILITY
Feature importance was used to identify which input
features contributed most to the model's predictions.

6. RESPONSIBLE AI
```

```
Model performance should be evaluated beyond accuracy.
Prediction errors, potential biases, and limitations
must be considered before real-world use.

7. LIMITATIONS
This project uses a built-in dataset and a single
train-test split. It does not establish fairness across
demographic groups or guarantee reliable predictions
on new populations.

8. ETHICAL CONSIDERATIONS
This model is an educational demonstration only.
It must not be used to diagnose or guide treatment
for any real person.

9. CONCLUSION
The project demonstrates model evaluation,
feature importance visualization, and basic responsible
AI considerations.
""")

print("=" * 55)
print("Report generated successfully!")
```

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       RESPONSIBLE AI & MODEL INTERPRETATION
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